Unit 13, Lesson 6: Comparing Data Sets and Graphs

Benchmark

MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.

Learning Target

- ☐ I can compare and interpret a numerical representation of data of one population and a graphical representation of another population.

13.6.1 Warm-Up: Would You Rather?

1. Would you rather…

   A. Pay for 3 gifts and get 1 of equal or lesser value free?
   B. Pay for 4 gifts using a coupon for 20% off?

   Explain your reasoning.

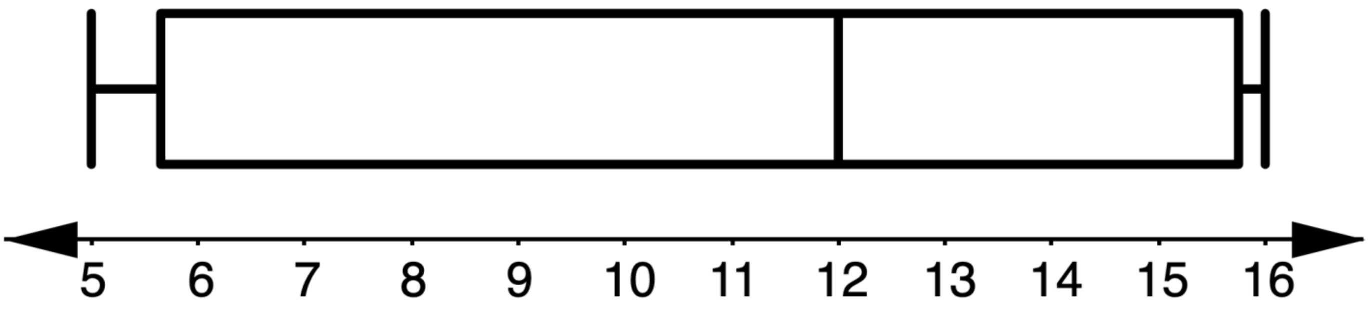
13.6.2 Guided Practice: Summarizing Data

1. The commute time, in minutes, for 24 randomly selected people in Duval County, Florida is shown.

9	10	15	17	18	20	20	21	22	23	24	25
25	25	25	26	26	29	29	29	32	34	38	40

The box plot displays the commute time, in minutes, for 24 randomly selected people in Union County, Florida.

Travel Time in Union County, Florida



Time (in minutes)

a. Explain why it would be beneficial to create a box plot of the Duval County data when comparing these data sets.

b. What are some measures used to describe data that cannot be determined from a box plot?

c. Use at least three terms from the term bank below to write a three-sentence summary comparing

travel times to work in Duval County to travel times to work in Union County.

mean	median	range	IQR
center	variability	upper quartile	
lower quartile	sample	population	

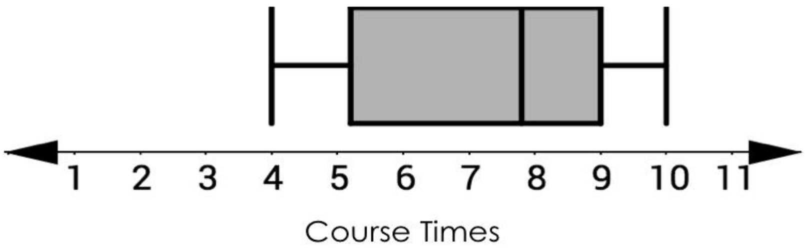
13.6.3 Try It: Comparing Data Sets and Graphs

1. The Deltona High School Wolves and Wildwood High School Wildcats cross-country teams ran an obstacle course. Each team's times are summarized.

Deltona HS Wolves' Obstacle Course Times

4:25	4:43	4:49	5:02	5:12
5:21	5:31	5:32	5:37	5:52
5:54	6:08	6:20	6:26	6:33
6:48	6:53	7:16	7:23	8:05

Wildwood HS Wildcats' Obstacle Course Times



a. Which statements are true about the data for the Deltona Wolves and the Wildwood Wildcats? Select all that apply.

- ☐ The median time of the Deltona Wolves is less than the median time of the Wildwood Wildcats.
- ☐ The fastest 25% of athletes on both teams complete the obstacle course in about the same amount of time.
- ☐ The interquartile range of the Deltona Wolves is less than the interquartile range of the Wildwood Wildcats.
- ☐ Approximately 50% of Wildwood Wildcats have times between 5 and 6 minutes.

b. Which team do you think is more likely to win the next race? Explain your reasoning.

13.6.4 Activity: Roll the Dice

With your partner, decide which of you will be "Partner 1" and "Partner 2." Partner 1 will be creating a line plot and Partner 2 will be recording a data set.

Using a standard six-sided die, each person will roll one die 25 times. After each roll, record the results in either a line plot or numerical data set below, depending on if you are Partner 1 or Partner 2.

1. When both partners are finished, copy their data below.

Line plot (Partner 1's rolls):

Data set (Partner 2's rolls):

2. Complete the table by calculating the measures of center and spread for your data. Share your data with your partner and add their data to the table.

	Partner 1	**Partner 2**
Mean		
Median		
Range		
IQR		

3. With your partner, discuss similarities and differences in your data. Summarize your discussion in the table.

Similarities	**Differences**

13.6.5 Your Turn!

1. A group of ten rowers that practice outdoors on the water (Group 1) are challenging a group of ten who use indoor rowing machines (Group 2). They want to see who can row faster using rowing machines.

   The time, in minutes, it took each of the ten rowers in Group 1 to row 500 meters on the rowing machine is represented in the stem-and-leaf plot.

   Group 1: Outdoor Rower Times (in minutes)

Stem	Leaf
15	$2, 8$
16	8
17	$0, 5, 8$
18	$0, 2, 3, 7$

   Key: $15|2 = 1.52$

   The time, in minutes, it took each of the ten rowers in Group 2 to row 500 meters on the rowing machine is represented in the data set.

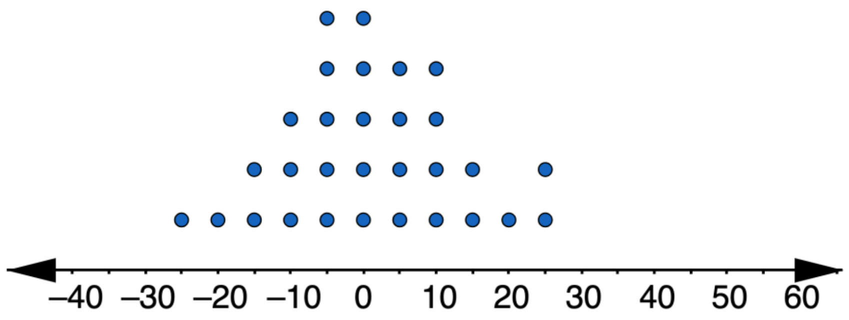
   $$\{1.63, 1.82, 1.95, 1.97\ 2.0, 2.02, 2.12, 2.20, 2.25, 2.88\}$$

For each statement, state whether it is true or false. For any statements that are true, explain why.

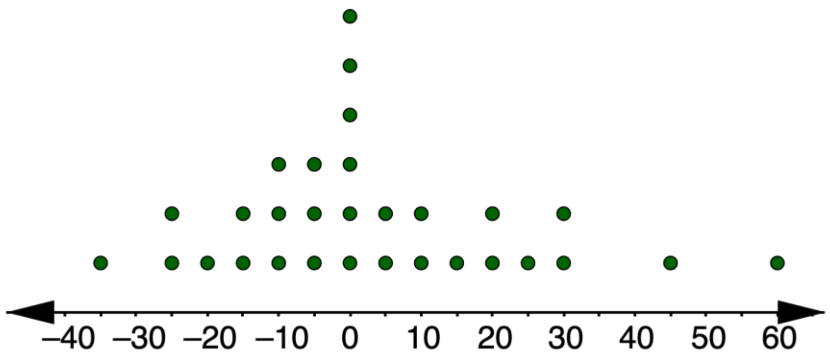
Statement	True or False?	Explanation
The center of the Group 1 times is similar to the center of the Group 2 times.		
The variability of the Group 1 times is similar to the variability of the Group 2 times.		
The Group 1 rowers that practice on the water are typically faster than the Group 2 rowers that row on machines.		
The Group 1 rowers that practice on the water are more consistent than the Group 2 rowers that row on machines.		

2. The dot plots display the arrival times of 30 flights for two different airlines.

Arrival Times of Airline P



Arrival Times of Airline Q



- A negative number represents the number of minutes the flight arrived before its scheduled time.
- A positive number represents the number of minutes the flight arrived after its scheduled time.
- A zero indicates that the flight arrived at its scheduled time.

Based on the data, from which airline would you choose to buy your ticket? Use measure(s) of center and variability to justify your choice.

13.6.6 Wrap-Up: Cool Down

1. Write a short note to a classmate who missed class summarizing today's lesson.

